Well-Worn but Unpaved: A Crosswalk Between Steam Tags and the Video Game Metadata Schema

Introduction

As a lifelong gamer, when I want to browse for new games to play on PC, I open Steam (the largest PC games digital distribution platform) and search by Steam tag, the folksonomy used by gamers to describe video games. I'm often searching by hyper-specific terms like "Metroidvania," "Souls-like," or "Tactical RPG." Any gamer who enjoys these types of games will understand exactly what type of game they describe. As described in Windleharth et al.'s 2016 article "Full Steam Ahead: A Conceptual Analysis of User-Supplied Tags on Steam," tags like these are "desire lines," or well-worn but unpaved paths that smart designers eventually pave by adding them to "the recognized system of walkways."[1] The University of Washington GAMER Group's Video Game Metadata Schema (VGMS) functions as one such system of walkways by creating the controlled vocabulary that institutional catalogers can use to describe games.[2] When Windleharth et al. compared the VGMS to the Steam folksonomy, they pulled 294 tags from Steam's popular tags list and identified gaps where community language had outpaced the controlled vocabulary. A decade later, Steam's populartags endpoint returns 446 tags. The folksonomy has only grown.

The artifact which accompanies this paper is a metadata crosswalk between the current VGMS (Gameplay Genre v1.3 and Mechanics v1.1) and the 61 Steam tags that SteamDB, a third-party site that displays and extends Steam's tag categories, lists under "Sub-Genres." The crosswalk itself lives in a public GitHub repo as structured JSON and a human-readable summary;[3] in this paper, I describe its scope, the methodological choices behind it, and what it reveals about the gap between community language and the controlled vocabulary. The crosswalk serves as an intervention in the datafication process this paper describes by forming a structured, repeatable, machine-readable artifact that makes gaps visible and actionable. What I find is that only 15% of those 61 tags map exactly to a VGMS genre term. Most strikingly, "Metroidvania," one of the tags I used as an example above, was specifically flagged by Windleharth et al. as a gap in 2016. Today, VGMS still has no preferred term for it. The desire line is well worn, yet still unpaved.

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[1] Travis W. Windleharth et al., "Full Steam Ahead: A Conceptual Analysis of User-Supplied Tags on Steam," *Cataloging & Classification Quarterly* (New York) 54, no. 7 (2016): 421–22, https://doi.org/10.1080/01639374.2016.1190951.
[2] Jin Ha Lee et al., "Video Game Metadata Schema," December 12, 2024, https://github.com/uwgamergroup/video-game-metadata-schema/blob/main/VGMS_v4.2_20241212.pdf.
[3] Lucas McGill, *JankMajesty/Steam-Tags-Crosswalk*, Python, April 29, 2026, released April 29, 2026, https://github.com/JankMajesty/steam-tags-crosswalk.

Why Video Game Genre Classifications Fail

So, what makes video game genre so tough to classify? In their 2017 article "Why Video Game Genres Fail: A Classificatory Analysis," Clarke et al. argue "that genres fail on a practical level because they do not adequately identify, collocate, and support retrieval of games for interested users."[4] These functions serve as the cornerstone of classification in library and information science. A controlled vocabulary must support these functions to make it useful.

The problem is that video games as a medium "are more than their narrative components: video games are interactive processes."[5] What's being classified is the experience of play as opposed to a book or film which inhabit one or two fixed narrative genres often involving no interactive component. Games are multi-dimensional which makes traditional classification awkward. Clarke et al. cite Wittgenstein who explains that "there are in fact no common properties shared by all games, but the category of games is united by family resemblance."[6]

This is true within genres as well. Exploration is seen as a defining feature of the Role-Playing genre of video games, but as Clarke et al. point out, "Fire Emblem games are commonly considered strategy RPG games, but they have no exploration component."[7] "Tactical RPG", the Steam tag that describes games like Fire Emblem, describes games like this by their collection of features, not by satisfying every condition of "Role-Playing" or "Strategy."

For video game classification, Clarke et al. recommend a faceted approach. By decomposing genre into multiple independent dimensions, hybrid games can be described as a combination of facets rather than forced into a single hierarchical slot.[8] VGMS at the schema level already implements this approach. The current release (v4.2) includes separate controlled vocabularies for Gameplay Genre, Mechanics, Mood, Theme, Setting and many other facets of game description. VGMS also marks Gameplay Genre and Mechanic as repeatable elements, so a cataloger can apply multiple terms from each vocabulary to a single game without forcing a choice.[9] The faceted structure is there at both the schema level and the cataloging level. What VGMS does not have is compound terms. "Tactical RPG" is not a concept in its own right within the Gameplay Genre vocabulary. Steam users have created compound labels (Tactical RPG, Roguelike Deckbuilder) and tags that cut across facets (Choose Your Own Adventure, Positioning)

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[4] Rachel Ivy Clarke et al., "Why Video Game Genres Fail: A Classificatory Analysis," *Games and Culture* 12, no. 5 (2017): 446, https://doi.org/10.1177/1555412015591900.
[5] Clarke et al., "Why Video Game Genres Fail," 446.
[6] Clarke et al., "Why Video Game Genres Fail," 457.
[7] Clarke et al., "Why Video Game Genres Fail," 458.
[8] Clarke et al., "Why Video Game Genres Fail," 458–59.
[9] Lee et al., "Video Game Metadata Schema."

that VGMS would express using multiple separate terms. The crosswalk operates at the tag-to-term level, which is where these gaps become evident.

Clarke et al. conclude their argument by expressing that "genre classifications and labels are as much 'bottom-up' products of a community as they are 'top-down' taxonomies."[10] The crosswalk aims to determine which community terms can be readily mapped to the VGMS and which community vocabularies are absent or tough to map so the gap can begin to be bridged.

Chains of Datafication

Steam tags don't appear out of thin air, and neither do VGMS terms. Both come from datafication processes that boil games down to their essential and accidental characteristics.[11] The process is constructed through layers of datafication initialized by the game developer and described and datafied through inscription by Steam users, SteamDB, and the VGMS maintainers (see Figure 1). Game developers create games. Steam users and developers apply tags. Steam requires developers to select at least five tags when submitting a game.[12] Steam users add more from there. Users can generate new tags, but Steam moderates the pool downstream, removing offensive or misleading tags in the process. The folksonomy permits users the freedom to generate new tags, but Steam ultimately decides what sticks. Steam itself maintains tags within editorialized categories (Genres, Sub-Genres, Themes & Moods, etc.), visible to developers via the Steamworks Tag Wizard. SteamDB then displays and extends those categories as a third-party classificatory system of record (CSoR) producing system-dependent truths that users and researchers (like myself, when I thought SteamDB was displaying official Steam classifications) treat as fact.[13] VGMS is the other CSoR in the chain, but one which has been thoughtfully designed and peer-reviewed.[14] VGMS is the product of a datafication process with the explicit goal of supporting video game cataloging in libraries, museums, archives, and other cultural heritage institutions where games are increasingly part of the collection.

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[10] Clarke et al., "Why Video Game Genres Fail," 460.
[11] Melanie Feinberg, *Everyday Adventures with Unruly Data* (The MIT Press, 2022), 67–68, https://doi.org/10.7551/mitpress/14198.001.0001.
[12] Steam, "Steam Tags (Steamworks Documentation)," accessed April 27, 2026, https://partner.steamgames.com/doc/store/tags.
[13] Beth Patin et al., "Toward an Etiology of Harm for Knowledge Organization: Onto-Epistemic Injustice in Classificatory Systems of Record," *KNOWLEDGE ORGANIZATION* 51, no. 7 (2024): 495–513, https://doi.org/10.5771/0943-7444-2024-7-495; Elliott Hauser, "Facts in the Machine: Systems of Record and the Performance of Sociotechnical Truth," *Journal of the Association for Information Science and Technology* 76, no. 2 (2025): 447–59, https://doi.org/10.1002/asi.24820.
[14] Jin Ha Lee et al., "Facet Analysis of Video Game Genres," paper presented at iConference 2014 Proceedings: Breaking Down Walls. Culture - Context - Computing, *iConference 2014 Proceedings*, March 1, 2014, https://doi.org/10.9776/14057.

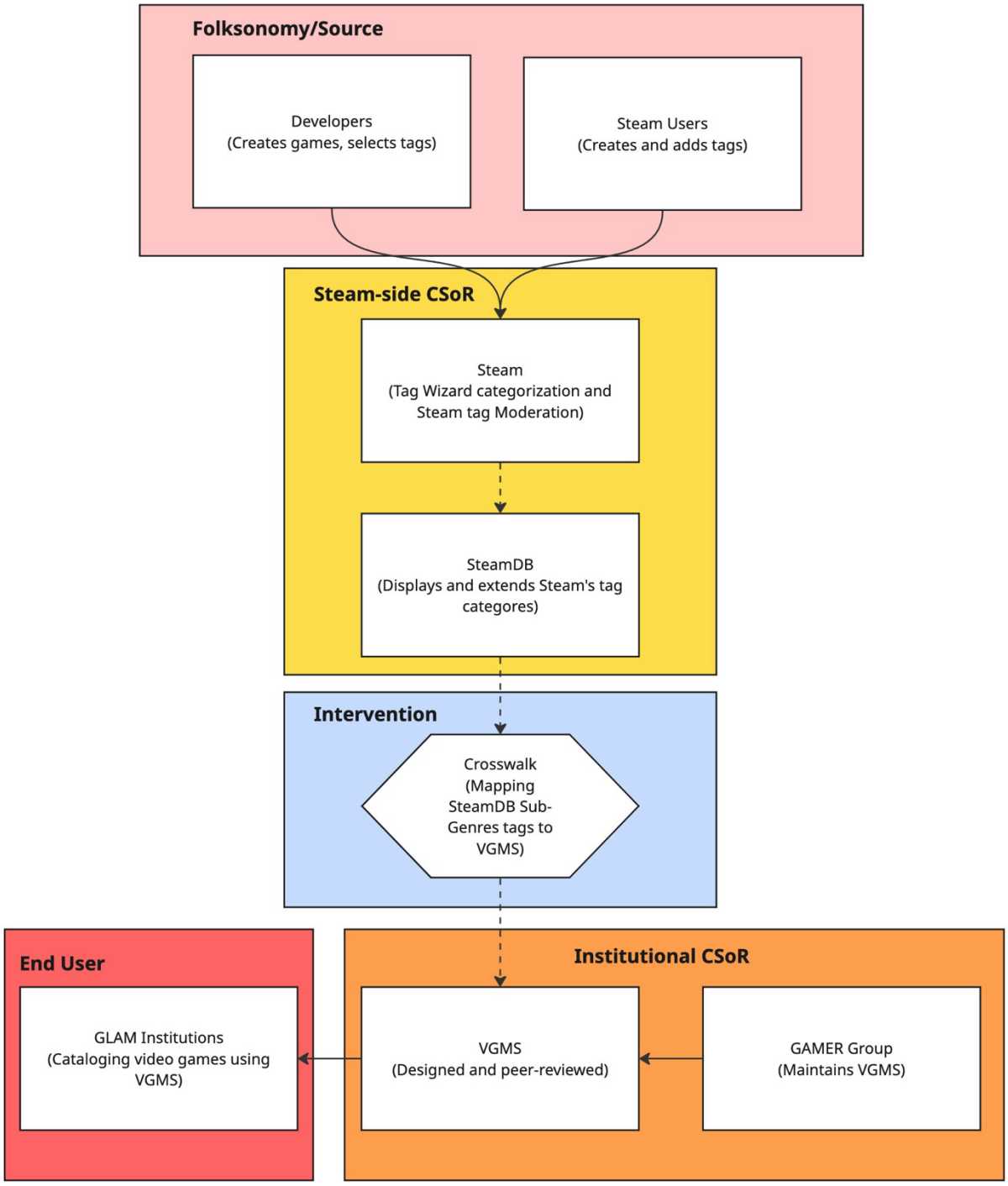

Figure 1. Datafication chain from Steam tags to GLAM cataloging. The dashed lines show interpretation without ownership. SteamDB displays and extends Steam's pre-existing classifications. The crosswalk supplements VGMS without being a part of GAMER Group.

Each layer involves what Hauser describes as "scientific forgetting," a process in which scientific fact is made certain through the act of forgetting complexity.[15] The Steam tag folksonomy preserves the messiness of user-generated description but forgets nothing, leaving the data unstructured. SteamDB forgets the messiness of Steam tags to gain a category structure by classifying them into boxes in which Steam tags don't always fit cleanly, while VGMS gains coherence by forgetting community terms which haven't yet been formalized into the controlled vocabulary.

The crosswalk sits between SteamDB and VGMS as a supplement that maps the 61 Steam tags SteamDB groups under "Sub-Genres" against two of the VGMS's many controlled vocabularies (Gameplay Genre and Mechanics). Windleharth et al.'s 2016 comparison was conducted from inside the GAMER Group;[16] what this project contributes is an external, reproducible artifact for periodic revision work that doesn't require permission from the schema's maintainers. Three users stand to benefit: a person searching for "Metroidvania" in a VGMS-based catalog gets a documented partial match to "Platform" instead of nothing; a cataloger describing a game like Slay the Spire see how "Roguelike Deckbuilder" decomposes into VGMS terms; a VGMS maintainer gets a popularity-ranked list of unmapped community terms.

Findings

Of the 61 SteamDB sub-genre tags I mapped, only 9 (14.8%) were an exact match to a VGMS Gameplay Genre term. 44 were partial matches and 8 had gaps where the Steam tag has no VGMS equivalent. Mechanics fared better: 16 exact matches, 31 partial, 14 gaps, though VGMS Mechanics includes some overly broad terms like "Choosing" which includes choices made in a Choose Your Own Adventure game like Mass Effect, or choosing answers in a trivia game like You Don't Know Jack. The Steam community has produced narrower subtypes, especially within shooters (Boomer Shooter, Hero Shooter, Looter Shooter), where the closest VGMS terms are Shooter and First Person Shooter. Tags like "FPS," "Match 3," and "Hack and Slash" map cleanly to existing VGMS terms, but they are the exception. Mechanics map more cleanly than Gameplay Genre, but only because Mechanics terms are overly broad and polysemous. The folksonomy's compound terms are harder to map to Gameplay Genre because they span boundary lines.

The partial-match column reveals the gaps. For instance, "Tactical RPG" is not included as a Gameplay Genre in VGMS. The closest VGMS terms are "Role-Playing" and "Tactics," and a cataloger describing a game like Fire Emblem in a VGMS-based catalog would apply both. What is missing from VGMS are compound tags that Steam users

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[15] Elliott Hauser, "What Is the Thermal Conductivity of Copper? The Production of a Fact Through Scientific Forgetting," *Proceedings of the Association for Information Science and Technology* 58, no. 1 (2021): 185–95, https://doi.org/10.1002/pra2.447.
[16] Windleharth et al., "Full Steam Ahead."

have nimbly employed. The same is true for "Roguelike Deckbuilder," where a game like Slay the Spire could possibly be cataloged under Card Game and Roguelike for Gameplay Genre and with Deck Building as the mechanic. However, Roguelike is a narrower term for Role-Playing in VGMS and Roguelike Deckbuilders typically forgo the Role-Playing elements of the Roguelike genre in favor of the other defining elements of the Roguelike genre such as perma-death and procedural generation. On review, I classified 17 of the 44 partial-match genre mappings and 24 of the 31 partial-match mechanics mappings as "related" rather than "narrower." "Narrower" belongs to true subtypes that narrow the scope a Gameplay Genre term (e.g. Shooter > First Person Shooter). "Related" better captures hybrids that combine multiple VGMS terms. The folksonomy's most distinctive advantage is not creating narrower subtypes, it is creating compound terms that span across multiple classificatory boundaries.

Another cluster of tags doesn't map cleanly to either Gameplay Genre or Mechanics and moving them into another VGMS controlled vocabulary doesn't help. For instance, "Choose Your Own Adventure" partially matches the Mechanic "Choosing," but as mentioned previously, the "Choosing" mechanic is overly broad as the VGMS game examples for this term refer to both choosing branching dialogue in Mass Effect and choosing answers in a trivia game like You Don't Know Jack. Two wildly different player experiences are flattened into one term. Mechanic terms like "Positioning" have the same issue. The Mechanic describes the type of positioning a player would encounter in tactics games like Fire Emblem and within puzzle games like Sudoku. Four tactics tags (Real Time Tactics, Turn-Based Tactics, Tactical RPG, Wargame) plus Sokoban (a puzzle game) all partially match Positioning, with no granular term to separate positioning in a tactics game versus a puzzle game. Other tags in this cluster (Exploration, Time Management, Runner, Collectathon, Typing, Spelling) describe gameplay concepts that VGMS either has no term for or the VGMS term is too broad to cover the distinction the community is making. The shortcomings of the CSoR are apparent here. Exploration, for instance, doesn't correlate to a sub-genre; it is more of a design aspect than a genre, and Steam's Tag Wizard doesn't classify it this way either; SteamDB does. Treating these tags all as sub-genres is a categorical claim that the data doesn't support.

Metroidvania is the Steam tag missing from the VGMS controlled vocabulary that inspired the crosswalk. The Metroidvania case makes the scientific forgetting process awfully noticeable.[17] VGMS draws its criteria for inclusion from the facet analysis that Lee et al. developed which grounds the schema in user warrant based on a "term's popularity and potential user familiarity."[18] Two years later, UW GAMER Group

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[17] Hauser, "What Is the Thermal Conductivity of Copper?"
[18] Lee et al., "Facet Analysis of Video Game Genres," 129.

scholars co-wrote the Windleharth et al. paper flagging Metroidvania as a gap.[19] Metroidvania has both the popularity and long-standing community recognition and familiarity that Lee et al. specify. The term's lack of inclusion raises questions about how warrant is applied in practice. Upon review, four community-recognized sub-genres warrant promotion to preferred terms within Gameplay Genre based on popularity and long-standing use within the Steam tag vernacular: Metroidvania, Souls-like, Roguelite, Boomer Shooter. Two more tags worth consideration, Clicker and Idler, are near-synonyms describing a genre of game defined by incremental progression through minimal player interaction. The remaining gaps from SteamDB's sub-genre tags (Exploration, Time Management, Runner, Collectathon, Typing, Spelling) are mostly cross-facet tags rather than candidates for new genres. However, caution is also warranted when choosing Steam tags for preferred term promotion. Adding trending terms which have no long-standing tradition clutters the controlled vocabulary, but failing to add widely used community terms fails the user and the researcher. Without consistent upkeep of the schema, a player searching for "Metroidvania" in a VGMS-based catalog will receive nothing in return. With the crosswalk, the player at least receives a partial match to Platform.

Conclusion

The crosswalk isn't an enduring panacea to bridging the gaps within the VGMS, and that's not the intention either. The intention is to make gaps visible and actionable: an archivist deciding which Steam tags to include in a finding aid or a cataloger describing a Roguelike Deckbuilder can use the crosswalk as a map between community language and the controlled vocabulary. VGMS maintainers can use it as a structured, popularity-ranked input for revision work.

Any specific mapping in this crosswalk will go stale, that is the nature of mapping folksonomies to a standardized schema. Steam users add tags faster than the VGMS can keep up. The crosswalk pipeline and resulting summary serve as a supplement to the work the VGMS maintainers do. The repository contains the scripts, the manual classifications, and the matching logic in a form that can be re-run against fresh Steam API data with a single command. The current snapshot provides actionable data today, the process provides a tool for revision in the future.

A VGMS maintainer reviewing the next crosswalk run could address gaps in four steps:

- Sort partial matches and gaps by popularity
- Evaluate warrant for each candidate term
- Draft scope notes and broader, narrower, and related relationships for promoted terms

[19] Windleharth et al., "Full Steam Ahead."

- Publish a revised version of the controlled vocabulary

The crosswalk contributes a reproducible, external input to the process. One issue encountered that this crosswalk doesn't fully solve is that the VGMS only distributes its controlled vocabulary as PDF. A readily available machine-readable version would allow crosswalks like this one to rebuild themselves automatically when VGMS updates.

The crosswalk reveals the desire lines Steam users have been etching for years. Metroidvania was flagged a decade ago yet remains unpaved. The crosswalk illuminates the path, VGMS has the tools to pave it.

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